Raviteja Velanati

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PROFESSIONAL SUMMARY

AI/ML Platform Engineer specializing in LLM evaluation, agentic AI testing infrastructure, and Trust & Safety workflows for Google AI (via GlobalLogic). Built 0-to-1 developer tools, full-stack sandbox environments, synthetic data pipelines, and human-in-the-loop evaluation platforms using Python, TypeScript, React, Angular, and C++. Ranked #1 of 11 engineers in platform throughput. Background spans fintech/enterprise backend (.NET), game development (Unity/C#), and production LLM serving. Focused on making LLM systems testable, traceable, and reliable at scale.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, C#, Go, Java, SQL, Bash/Shell, HTML5/CSS3

Frameworks & Libraries: React.js, Angular, Node.js, FastAPI, Django, .NET Core, LangGraph, LangChain, PyTorch, Playwright, Apache Spark/PySpark, Unity

AI & LLM: LLM Platforms, RAG Systems, Agentic Orchestration, Prompt Engineering (CoT, persona, adversarial), promptfoo, MLflow, Autogen, AI Safety & Adversarial Testing, Evaluation Frameworks, Synthetic Data Pipelines, Human-in-the-Loop (HITL) Systems

Cloud & DevOps: GCP (Cloud Storage, BigQuery), AWS (EC2, ECS, S3, Lambda), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD, Linux Administration, Internal Developer Platforms, Simulation Sandboxes

Tools & Practices: Google3 toolchain (Cider, Piper, CitC, Blaze, Critique), textproto, Google Apps Script, Cursor, Claude Code, Gemini, Agile

Data: BigQuery, SQL, Neo4j, Multi-modal data pipelines (video, audio, OCR, text)

PROFESSIONAL EXPERIENCE

GlobalLogic (Client: Google AI) Santa Clara, CA | Oct 2024 – Present

Software Engineer – AI Platforms, Infrastructure & Safety Testing. Progressed through three roles supporting Google AI across LLM evaluation, Trust & Safety, and Responsible AI.

Senior AI Solutions Engineer – Responsible AI & Novel AI Testing Nov 2025 – Present

- Designed, built, and deployed **13 full-stack sandboxed web applications** (e-commerce, grocery, food delivery, fintech, messaging) from scratch to safely reproduce edge cases and evaluate autonomous browser AI agents (TypeScript, Angular, Python, Apps Script).
- Architected a synthetic data seeding framework across **48 multi-modal user personas** (healthcare, wellness, youth safety, enterprise) and **9 cloud services**, injecting longitudinal user histories (Email, Conversational AI, Search, Video, Photos, Drive, Calendar, Notes, Tasks, Fitbit).
- Built an automated Python Data Integrity QA suite auditing post-injection state across all 9 services, guaranteeing **100% numerical accuracy** between source datasets and target environments; added schema validation, preflight checks, and audit logging.
- Built Python/Playwright browser automation to bypass manual login across all 48 test accounts, automatically enabling biometric face clustering and identity labeling in Cloud Photos.
- Owned a multi-language video segmentation and cloud storage pipeline across 18 workflows, refactoring the media slicing engine to process **1,800+ recordings into 40,000+ two-minute clips across 8 languages** and cutting processing time by **10x**.
- Resolved a critical data-injection blocker in **10 days** by re-engineering bulk extraction and mail-flow ingestion pipelines with structured diagnostic logging, unblocking 3+ dependent engineering teams.
- Built full-stack telemetry dashboards (React, TypeScript, Node.js) for real-time execution tracing of production AI agents, reducing mean-time-to-diagnosis by **~60%**.
- Engineered warm-pool Docker/Kubernetes infrastructure that cut sandbox initialization latency by **70%**, accelerating daily deployment velocity for core platform teams.
- Built async Python data pipelines for multi-modal datasets at scale and CI/CD regression harnesses that prevent capability drift across model weight updates.

Senior Software Developer – Plugin & Data Solutions Architect (CrowdCompute) Aug 2025 – Sep 2026

- **Ranked #1 of 11 developers** in platform throughput, delivering **2,000+ ingestion and extraction batches across 460+ evaluation workflows** with zero queue backlog, including weekend live-event operations.
- Designed and deployed progressive layout containers and textproto configurations for Side-by-Side (SxS) human rating interfaces across **19 custom plugin builds**, supporting multi-model LLM comparisons, multilingual query vetting (EMEA, APAC, LATAM), indirect prompt injection (IPI) evaluations, and professional-writing benchmarks.
- Built React/TypeScript annotation templates and modular automation plugins for human-in-the-loop validation, connecting user-facing interfaces to distributed cloud clusters and enabling high-quality human-labeled training data.
- Owned Python ingestion, extraction, and Stage 1→Stage 2 deduplication scripts across **80+ multi-stage workflows** (200+ deliveries, including 55 P0 requests); prevented 60-day server expiration timeouts, fixed payload serialization overflows, and mentored junior developers.
- Onboarded multi-modal datasets (Zip/HTML Chain-of-Thought trajectories, streaming video, OCR, audio) and built automated QC recovery pipelines; managed credential governance, API proxying, and GCS access controls.

- Developed domain-specific evidence retrieval plugins and pipelines in Python, TypeScript, and Go within the Google3 development loop.

Python & C++ Developer / Prompt Engineer Nov 2024 – Aug 2025

- Built and maintained Python/C++ backend services to host and serve LLMs with low latency and high throughput under heavy production query loads.
- Crafted system and evaluation prompts (Chain-of-Thought, persona role-play, adversarial stressors) that significantly improved LLM pass rates and instruction adherence.
- Ran LLM robustness evaluations to detect prompt injection and instruction-contradiction vulnerabilities using objective automated assertions.
- Engineered a custom **YAML Editor** to validate promptfoo configurations and a **Promptfoo Analyzer** to aggregate results and drive A/B prompt optimization in CI/CD pipelines.

Additional Google AI Workstreams – Trust & Safety, Novel Testing, Google Meet Fallback Diagnostics Dec 2025 – Present

- Created adversarial evaluation frameworks and standardized quantitative scoring for multi-turn agentic task completion, identifying **40+ edge cases** in production Gemini models before deployment.
- Designed diagnostic test suites and PyTorch validation pipelines for Gemini fallback handling, increasing production issue resolution speed by **50%**; integrated LangGraph-based tooling that translates natural-language specs into structured test sequences within enterprise CI/CD.

MassDigi at WPI, Worcester, MA – Software Engineer & Game Developer May 2023 – Aug 2023

- Programmed AI-driven NPC behavior in Unity/C# and C++ using finite state machines and behavior trees; built in-editor debugging and logging tools to reproduce AI anomalies faster.
- Optimized memory allocation, asset loading, and object pooling to sustain **60+ FPS** under heavy rendering loads; resolved **30+ integration defects** in agile sprints and shipped mechanics on schedule.

Wipro, Hyderabad, India – Software Engineering Intern (.NET Full-Stack) Mar 2022 – Aug 2022

- Developed web applications and backend services on the .NET full-stack ecosystem (C#, SQL); accelerated C# architecture proficiency through structured mentoring and passed technical assessments.

SELECTED PROJECTS

Agentic AI Evaluation & Trajectory Analysis Platform

Python, LangGraph, LangChain, FastAPI, Docker, Kubernetes, RAG

- Architected an evaluation platform processing **10K+ test cases daily** to benchmark production LLM and multi-agent behaviors across containerized sandboxes; prevented capability drift across **12+ model iterations**.
- Implemented automated retry logic and outcome-analysis pipelines parsing tool-call sequences and reasoning traces, eliminating **95%+ of silent failures** in edge-case handling.

AI-Native Developer Tooling & Automated Test Harness

Python, React.js, TypeScript, FastAPI, Docker, AWS Lambda, CI/CD

- Built an internal backend service translating natural-language product specs into structured coding sequences and CI/CD test workflows; reduced feature-to-production time by **40%**.
- Integrated with enterprise deployment pipelines with strict output parsing, API-first access, and security guardrails; shipped **30+ automation plugins** and eliminated manual testing overhead across **8+ internal teams**.

EDUCATION

Clark University – M.S., Computer Science (GPA 3.5/4.0) *Worcester, MA | Jan 2023 – Dec 2024*

Applied Data Analytics, Software Engineering, Design & Analysis of Algorithms, Data Visualization, Cyber Security Fundamentals, Network Architecture & Design, Capstone

Malla Reddy College of Engineering & Management Sciences – B.Tech, CS (GPA 3.6/4.0) *Aug 2018 – May 2022*

CERTIFICATIONS & ACHIEVEMENTS

- **Ranked #1 of 11 engineers** in platform throughput (Google AI / GlobalLogic)
- Google Gemini for Developers • Microsoft Azure AI Essentials (Microsoft & LinkedIn) • Pair Programming with AI • Walmart USA – Advanced Software Engineering • Datacom – Software Development